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COMPUTER GRAPHICS

INDEPENDENT STUDY

Technology for Automatic Sign Language Interpreter

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1 Introduction

An extensive research was performed to gather all information regarding the most influential and important events that have transpired throughout the development and evolution of sign language automatic translators. Many important events, or reports, were found during the investigational process of this project, but not all of them were explained in depth. The reason for this is because there was not enough time to thoroughly explain them. I carefully selected the most relevant events from these projects, which I will delve into exhaustively. In this manner, the most influential research conducted on this topic will be discussed. Based on my research, the origin of automatic translators came from the need of wanting to fix, or lessen, the communication gap between signers and speakers. Most signers tend to be deaf and hard of hearing; hence, these individuals depend on sign language to communicate with each other, and human interpreters to communicate with speakers. Human sign language interpreters relay concepts and ideas between spoken and sign languages, such as English and American Sign Language (ASL), respectively. Therefore, they must thoroughly understand the subject matter in which they work to convert information from a spoken language to a sign language. This means that they must become familiar with the subject matter that the speaker(s) will discuss. Similarly, they must remain sensitive to the cultures associated with their language of expertise while combining languages, a technique that enables them to be fluent in at least two languages. Even though human interpreters have been an indispensable aid to help signers communicate with speakers, when compared with a sign language automatic translator, some deficits become apparent. First, there has been an explosion of people with hearing loss throughout the world. From this originates one of the disadvantages that deaf and hard of hearing people are facing today, a small number of human sign language interpreters. Human sign language interpreters are not accessible all the time, which bring up another issue faced by the deaf and hard of hearing community; availability. Throughout my research about the history and evolution of sign language automatic translators, it was proven that software tools used to translate sign languages to spoken languages yielded better results when improving a child's language literacy level. This brings up another issue; sign language human interpreters have limitations. With the help of automatic translators, the deaf and hard of hearing community have a better chance of achieving educational opportunities similar to their non-deaf counterparts. Sign language automatic translators would provide signers with something more than a tool; it would help them improve their education, create more jobs, improve their communication skills by using customizable system(s), and even improve their social lives.

1.1 Human Sign Language Interpreters

1.1.1 Preparation

In the early 1970's, interpreter training programs (ITP), that were generally six to eight weeks in duration, were established. These programs focused primarily on settings in which interpreters might expect to work such as religious, telephone, courtroom, medical, among others. The linguistic component of these early programs was on vocabulary building. There was limited discussion concerning emotional, physical, and mental processes involved because there was limited scholarly material available for such use[35]. Today, there are about thirty-eight colleges and universities nation-wide offering a BA in interpreting or deaf studies (with concentrations in interpreting)[5]. To become an interpreter, individuals must display bilingual and bi-cultural proficiency and have the ability to mediate meanings across languages and cultures simultaneously and consecutively. This process can take years of intensive practice and professional training because interpreters are highly recommended to complete an associate, bachelors, or master's degree from an interpreter-training program[35]. This is prior to obtaining a certification by the Registry of Interpreter for the Deaf (RID). The National Interpreter Certification (NIC) is a test administered by the RID and the National Association of the Deaf (NAD). Most graduates of the ITP need at least 2-5 years of interpreting experience before passing the NIC test. As of July 2012, a BA, in any discipline, is required to sit for the test. The degree or certificate can take a minimum of three years starting with ASL III. If starting with ASL I, it can take a minimum of four years to complete all ITP coursework[5]. Students may pursue the Associate in Science degree in American Sign Language (ASL) or the short certificate in Interpreting Training (ITP). Both programs are designed to prepare students to begin a career as sign language interpreters[36]. The Associate of Arts degree includes taking courses in the major, along with general education coursework. The Certificate of Achievement focuses on coursework in the major, and is generally for those who already have an AA or BA degree. Another type of assessment is the Educational Interpreter Performance Assessment (EIPA), an assessment taken by some wanting to do educational interpreting, which does not have a degree requirement at this time. RID offers alternate paths for those without a BA, such as having a certain number of college units, years of experience in full-time interpreting positions, and proof of continuing education[5].

These preparatory steps enable interpreters to learn specialized techniques and ethical considerations for interpreting in a variety of settings with people of various ages. In the case of an educational interpreter, the individual is faced with many responsibilities. The process of becoming fluent in ASL and taking the required courses and tests to become certified, applying specific techniques and practices in the classroom, and, above all, being an advocate for the Deaf student[35].

1.1.2 Sign Language Interpretation

Human sign language interpreters relay concepts and ideas between spoken and sign languages, such as English and American Sign Language (ASL). Therefore, they must thoroughly understand the subject matter in which they work to convert information from a spoken language to a sign language. This means that they must become familiar with the subject matter that the speaker(s) will discuss, a task that may involve research to create a list of common words and phrases associated with the topic. Similarly, they must remain sensitive to the cultures associated with their language of expertise while combining languages, a technique that enables them to be fluent in at

least two languages. Their primary language is their native speaking language, and their secondary language must be a sign language of which they must have nearly perfect knowledge. Translating between a spoken language and a sign language requires interpreters to pay attention carefully, and to understand what is being communicated in the hearing and signing audience. Therefore, every interpreter must express their thoughts and ideas clearly while having strong research and analytical skills, mental dexterity, and an exceptional memory[44]. These attributes help them to provide a common link between signing and the hearing audience. Interpreters also use a technique named scaffolding, which is when a interpreter provides necessary vocabulary, definitions, and background information to expand on previously learned concepts, new concepts and terminology. Furthermore, they must have signers feel comfortable within the hearing audience. Hence, they need to identify all speakers in the audience by using name or fingerspell the name of the person commenting rather than pointing. Sometimes, an interpreter must instruct the speaker to take turns when speaking so they can accurately convey the message to all signers. Interpreters must keep a direct interaction with all signers, and require the hearing audience to speak directly to the signer and not to the interpreter. In fact, interpreters must monitor a signer's understanding of the material or concepts being discussed on a continual basis. This means that they must meet one-on-one time with a signer to clarify messages by altering the interpretation to match the person's language level[35].

Sign Languages, in the same way as spoken languages, can be translated simultaneously and consecutively[15]. When translating simultaneously, the interpreter must listen and sign at the same time. Thus, the interpreter must begin to convey a sentence being spoken while the speaker is still talking. Ideally, simultaneous interpreters should be so familiar with the topic that they are able to anticipate the end of the speaker's sentence. And, when interpreters need a high degree of concentration they work in pairs with each having 20-30 minute segments. On the other hand, when translating consecutively, the interpreters begin signing only after the speaker has verbalized a group of words or sentences. These interpreters often take notes while listening to the speakers by implementing some type of note-taking or shorthand system. In this case, interpreters segments may vary in length, writing style, and subject matter. When translating a text into a sign language, interpreters usually read the entire text to get an idea of the subject and to identify any unfamiliar words. Sometimes, additional readings are needed before interpreters start and finalize their sign translation. In this way, they can consult the text's originator or agency to clarify unfamiliar ideas, words or acronyms[44].

Overall, interpreters must manipulate sentences and ideas to make them flow with the same coherence as in the provided source. They must bear in mind any cultural references that may need to be explained to the intended signing audience, such as colloquialisms, slang, and other expressions that cannot be translated. In other words, the translation must read as though it originated in the sign language of choice. Interpreters in the United States must be fluent in English and American Sign Language, which combines signing, finger spelling, and specific body language. Most sign language interpreters either translate, aiding communication between English and ASL, or transliterate, facilitating communication between English and contact signing, which is a signing that uses a more English language-based word order[44].

1.1.3 Disadvantages

Sign language interpreters are humans. Therefore, they will make mistakes one way or the other. The need for a perfect sign language translator comes from the need of using signing, not signing human interpreters, to be able to communicate as normal as possible. A three-way communication, which in this case means having a human sign language interpreter, will always involve issues related to the availability, privacy and comfort of the signer.

- **Availability:** Normal activities such as going to the doctor, attending to a professor office hours, among others basic activities need to be scheduled based on human interpreters availability. Interpreters are not always available, and part of this has to do with the fact that sometimes the demand exceeds the number of interpreters available in an institution. Also, in emergency cases such as accidents, legal matters, and unexpected cases the person might not be able to fully comprehend the signer. Therefore, the person would have to wait until an interpreter is available in order to understand the signer.
- **Privacy:** Signers do not have the privilege of being able to express their feelings to hearing friends without a three-way communication. Besides of the fact that interpreters must follow certain privacy protocols, we all know how important is for us to be understood without having to think what the other person thinks about us. Even though most interpreters strictly follow their privacy code, the feeling of lack of privacy will always be there. Therefore, the way signers express themselves towards a hearing person may not always be the way they feel.
- **Discomfort:** The lack of privacy can lead to anyone feeling uncomfortable about talking certain topics, especially if it is related to somebody's personal life. A signer might not be comfortable exposing personal matters in front of a third person, even if this person is the interpreter. For example, a signer that goes to a doctor for any personal reason will respond more comfortably if the doctor knows ASL, or has a way to understand ASL, than having an official interpreter.

These are some of the basic reasons to why interpreters might become a source of impotence sometimes, even though their role has been extremely important in the lives of not-speaking signers. The fact of the matter is that technology is evolving fast, and signers will soon be able to be independent from human interpreters to do things that any speaking person does. Things such as flying, driving, and walking to any destination alone. These are things that not-speaking signers could do if they had the ability to ask questions to any one, including hearing people. This is an ability that a normal human interpreter cannot provide not-speaking signers with.

2 Automatic Sign Language Translators (ASLT)

2.1 The Need of ASLT

The most basic needs for Automatic Sign Language Translators can be explained in the following five sections, “Improve English Literacy,” “Software Accessibility,” “Few Interpreters,” and “Increase of Hearing Loss.” Keep in mind that these are only some of the most important needs, there are many other needs that are not being addressed in this project. However, most of them depend on the sections below.

- **Improve English Literacy Level**

Without aural exposure to spoken English during the critical language acquisition years of childhood, many deaf adults have below-average levels of written English literacy. Studies have shown that the majority of deaf high school graduates in the US have only fourth grade English reading level. Many of these people with low levels of English literacy are actually fluent in American Sign Language (ASL). These low levels of literacy have become an even more significant issue in recent decades as new information and communications technologies have arisen that place an even greater premium on English literacy skills[17].

- **Software Accessibility** An issue that has become more significant in recent years is the accessibility of computer software and websites to people with disabilities. Unfortunately, few software companies have addressed the connection between deafness and literacy, and so few computer user-interfaces make sufficient accommodation for the deaf[17].

- **Small number of interpreters** As a minority community, with limited access to heard speech, deaf people rely on the relatively small numbers of hearing people who learn sign language to be their interpreters. Since sign languages have different grammatical structures to any spoken languages translation is a problem in most contexts without provision of highly trained interpreters[2].

- **The Increase of Hearing Loss** An estimated 360 million people worldwide suffer from hearing loss, including approximately one-third of all people over the age of 65. For people with profound hearing loss the primary form of communication is sign language[48].

An automatic translator would allow a signer to be independent from needing a human interpreter. It would give signers the opportunity to express themselves without having to worry about the availability of their translator. They would be able to have private conversations with their friends without having to worry what would the interpreter think about their thoughts or they say. And lastly, they would be able to disclose any personal information to their doctor, psychiatry, among others professionals, without having to feel uncomfortable about their exposure. In fact, an automatic translator would allow signers to live a normal life and do things that any hearing person does. Things such as traveling alone, socialize with anybody and anywhere, work in places where they constantly interact with hearing people. In other words, an automatic translator would not just give signers freedom, but it would increase their chances of improving their education, getting a job, and be more social.

2.2 Benefits of ASLT

Throughout my research I found many advances in this new technology that the not-speaking signing community would take advantage of. Also, I found the benefits that an ideal automatic sign language translator would provide to a signer with. This second one is important since it is the source of inspirations to the multiple attempts to build an automatic sign language translator, which has been broadcasted during the past few years.

1. Benefits of a New Technology

- **Improve Communication & Social Life:** The intended purpose of automatic sign-language interpreter is to enable new opportunities for communication between those conversant in Sign Language and those who are not[4].
- **Improve English Literacy:** The aim of the technology is to empower sign language users by enabling them to overcome the communication challenges they can experience through portable technology running on any operating system equipped with a standard webcam. The portability and flexibility of this system, together with multi-platform functionality and customizable software, carries a potentially positive impact for users in a number of face-to-face situations, such as school and employment[4].
- **Improve Education:** It would enable a more ubiquitous use of Sign Language interpretation when live interpreters may not be available. For example, in a classroom setting – the system would allow the provision of Sign Language via an on-screen avatar when a live Sign Language interpreter is not available for a lecture[4].
- **Increase Job Opportunities:** Another positive impact for users is finding employment and communicating more effectively on the job. That is, using a device that accurately and rapidly translates signs to text to enable a user with a hearing or speech disability to expand the number of jobs he or she applies for, because the communications barrier between signers and non-signers is reduced[4].
- **Satisfies the Outgrowing number of signers:** Systems can be customized for individual use, which means that any signer can create her/his own set of signs and gestures to associate to them with their own words and concepts. This would allow sign language users to actually develop their own signs for concepts and terms they need to have in their vocabulary, but they may not have been able to express easily when using a given sign language. Also, younger learners with speech disabilities can use the system to create libraries of customized hand gestures and signs that express “domain-specific concepts” needed to discuss topics of study with teachers and others. Furthermore, the customizable feature would enable those with more limited motion or other physical disabilities to create meaning-specific gestures tailored to physical capabilities. In fact, regional variations of culture and custom can be integrated with the basic vocabulary of the system enabling one’s language to be personalized for everyday situations[4].
- **Availability:** Although the sequence of signs from a camera can be displayed as text on

the same device it has been detected from, the sequence can also be transmitted and viewed remotely. Therefore, it is possible to have the sequence sent as an SMS message or as a Bluetooth command to control an appliance[4].

2. Ideal ASLT Benefits

From a signer's point of view, having the ability to be understood through portable automatic sign language interpreters could make them feel as follows:

- In Schools: There would be no need for oral schools or interpreters. All teachers, administrative staff, security, et al., would be able to communicate with their deaf or hard of hearing students[47].
- At Work: Deaf, or hard of hearing, coworkers could communicate with hearing coworkers[47].
- Family: Deaf, or hard of hearing, parents could communicate with hearing parents. Subsequently, hearing parents could communicate with their deaf or hard of hearing children[47].
- Legal Matters: If Deaf, or hard of hearing, people get in trouble with the authorities, they would be able to communicate with them. Also, deaf or hard of hearing prisoners would communicate with other hearing prisoners and would not be isolated anymore[47].
- In Politics: Deaf or hard of hearing people would be able to run on any level such as local, state, or federal. Also, they would be able to vote for candidates who actually understand sign language. In fact, the president of the United States would understand sign language in all of his speeches[47].
- Jobs: Deaf or hard of hearing actors would be able to work in television along with hearing actors without the need for captions[47].
- Divided Society: The concept of the meaning of "disability" would change for the deaf and hard of hearing community. They would not be considered "disabled" because they would not need interpreters or captions. In other words, the society would no longer be divided between speakers and signers[47].

2.3 Advance Throughout the Time

Technology is rapidly changing the way we see and do things, including communication. The barriers that exist between people with speech disability and hearing people are slowly diminishing. This section briefly describes the advances that automatic sign language translators have had throughout the years. These advances have been obtained through the implementation of multiple techniques from different disciplines of computer science. However, the advantages obtained and proposed by these techniques are going to be discussed from a computer graphics point of view. Some of the most successful techniques used for automatic translations have been implemented based on Graphics Tools, Natural Language Processing, Robotics, Glove Based Methods, Neural Networks, Hidden Markov Models, Vision Based Methods, Three-Dimensional (3D) Animation, and Computer Graphics.

2.3.1 Graphics Tools - Aided Instructions

These applications allow students to create their own work through sign-language related authoring software. Software such as Authorware gave them the ability to create their own avatar translations for inclusion in videos, web sites, and more. A rudimentary software program developed for sign language is available free of cost and allows users to create animated sequences in any sign language[34?]. Usability tests have been conducted to obtain early feedback on animated computer graphics to present translations of English into American Sign Language (ASL). These animations are part of the development of a personal digital translator for signers. ASL is a visual language, therefore it is important for the animations to be visually realistic, recognizable, and comprehensible to ensure that the components of the translator are usable by the deaf community[34?]. According to this article, the creation of fingerspelling prototype would be an opportunity to develop a scale model of the problem set to refine solutions that would apply to the animation of hand shapes used in ASL[7].

In 1977, a multimedia database tool named “SignStream Project” was created; containing digitized video data and a representation of the data in a linguistic format[22]. In 2000, a system named Translation from English to ASL by Machine (TEAM) studied computational linguistics, computer graphics, and autonomous agents that led to the development of sophisticated communicative agents over the past few years prior to 2000, bringing a new perspective to the machine translation research[34]. According to this article, the engineering of language-based smooth, expressive, natural-looking human gestures could give us useful insights into the design principles that have evolved in natural communication between people. Their system prototyped a machine translation system from English to ASL, taking into account the linguistic, visual and spatial information associated with ASL signs[51]. Another, ICICLE combines ICAI with another domain of artificial intelligence, an NLP. In this case, the NLP has a tougher job than with “regular grammar checks or translators.” It analyzes the English writing, while taking into consideration the typical errors present in deaf writing samples and thus inferring what the writer “meant”. According to this article, their system introduced assisted writing tools for deaf users of ASL. The novel aspect of this system was to view the task faced by writers as one of second language acquisition. The methodology was presented for modeling the second language acquisition process[26].

In 2001, another ICICLE implementation was presented as a methodology for the development of a user model for CALL. CALL is a system that captures various levels of language acquisition using individualized overlays supported with stereotypes. This implementation was focused on the empirical analysis of the order of written English grammatical structure acquisition in their learner population used to develop stereotype layers in their models. According to this article, the system implementation was based on a windows application with a text parser that uses an English grammar augmented by a bug catalog or “misrules” capturing typical errors made by the learner population. In this way, the system recognizes grammatical errors, delivering one to two sentence explanations of each error on request. If the system finds more than one possible analysis of the grammar structure, it chooses arbitrarily. The envisioned system was viewed as two interacting modules. The first module was the identification, used to analyze the text and identify errors. The second module was the response generation, used to generate and display the tutorial dialog. Both modules make use of knowledge bases, because this covers the general information to analyze and explain grammatical structures, and the information about the current user such as his level of knowledge, his reaction, and system efforts in the past [28]. This same year, the THETOS system was developed requiring input data in the form of a text file and providing its full linguistic analysis to produce the output in the form of an animated sequence[34]. This system is used in Poland in medical settings where the personnel type a question into the system to be analyzed as a phrase, and decide what animation should be created to render the same message in Polish sign language. The word has characteristics that when put together result in a sign by the system animation technique, “OpenX”[41][19]. Next, a computer-generated synthetic interpreter named Paula was developed by a team of faculty and students in the school of computer science, Telecommunications and information Systems at DePaul University in Chicago. The system used speech recognition and sophisticated animation to translate spoken English into ASL[37]. One of the first planned applications of Paula was airport security. The research team also wanted to use it to replace closed captions such as the Simon project described earlier[34?]. The first step in the Paula Project was to create a database of signs to be used as the lexical database for the translator. This project did not capture the model signs through virtual reality but took a different approach because motion-capture data was thought to be too inaccurate and is recorded in numerical terms that are hard to modify. Paula instead uses an animation software package that has been customized for sign transcription[34?]. Lastly, the team proceeded to build a dictionary of complete signs by selecting hand shapes and positions from a friendly menu-driven interface using the same animation software. The advantages of storing a language as ASL are the followings: First, the linguistic information enables researchers to make quick queries about lexical properties of the language. Second, the graphical information enables design programs to synthesize graphic animations of ASL signs. Third, the dictionary information provides an on-line resource for English-ASL translations[12].

In 2002, the ICICLE project was enabled to design a user model to incorporate a representation of the language acquisition process basing its design upon current research in language acquisition and in the acquisition cognitive skills[34?]. The target of this project was the native, or near-native, people of American sign language to provide them with assistance as second language learners of English. This same year, another implementation of ICICLE was presented and its focus was to develop a system that adapts to the learning strengths of the user. It had the flexibility to engage in multi-modal, multi-lingual tutorial instruction utilizing English and the native language of the user. According to this article, their implementation was applied based on second language acquisition and the analysis of their corpus of writing samples from their learner population towards establishing the relationships between features in the SLALOM model. This model was essential

for them to finalize their design and system implementation. The second module individually instructs the student on the appropriate grammar lesson. The goal is to use an avatar to explain in ASL to increase the effectiveness of the instruction[27]. By 2006, the first sign language authoring product was released. This product was named Sign Smith Studio, it had nonlinear editing tools with five different racks such as speech, gaze, expression, cosign, and sign. It had a dictionary of 2,000 signs and automatically translated from written English to Signed English. There were twelve different avatars to choose from, these were VRML based, providing students with a powerful technology[34?].

The role of sign dictionaries has been revolutionized by new computer-based sign language technologies. Currently, with cameras, virtual gloves, among other technologies the user can sign the word and have the computer find the definition with the aim to benefit students that know signs. Another application of this technology was the Relay Phone or Internet Service, whereby a hearing interpreter must serve as an intermediary during conversations between deaf and hearing persons. This third party is soon to be eliminated with the use of translating avatars in some situations. According to this article, in the field of educational interpreting, an avatar in cases could potentially interpret limited activities where qualified interpreters were not available.

2.3.2 Natural Language Processing

Before elaborating in the evolution history of the projects and systems that helped us to get to the stage in which automatic interpreters are nowadays, I will summarize some of most influential first developments that may have provided us with a broader path to get to where we currently are. First, Valerie Sutton invented “Sign Writing” in 1974[34?]. It contains over 600 symbols and can describe all parameters mentioned above. Sign Writing is useful for studying ASL grammar and is more “readable” to people than the other systems. The HamNoSys system was developed as a scientific and research tool in the 1980s[14]. It consists of 200 symbols covering the above-mentioned parameters. Transcriptions are precise but long and cumbersome to decipher, which is used by Tessa. Second, by 2002 the Szczepankowski System was developed. This method was used in the THETOS (Text into Sign Language Automatic Translator, which operates in Polish) project[11]. Lastly, by 2004 a free DOS-based software program was available to type the symbols[42]. By now, researchers were able to convert Sign Writing into SignWriting Markup Language and subsequently into virtual reality animation sequences using MPEG-4 coding techniques[33]. A web online tool that allows users to convert text to animated sequences rendered in an avatar of choice.

In 1965, Casterline, and Croneberg developed “Stokoe”. They proved that the components of ASL, fit together, could form a linguistic structure comparable with that of spoken language. The Hamburg Notation System for Sign Languages (HamNoSys) is an alphabetic system describing signs on a mostly phonetic level, and its first version was defined in 1984. The first published version Prillwitz et al., 1987 was designed to be usable in a variety of contexts. As many sign notation systems developed in the last 30 years, HamNoSys has its roots in the Stokoe notation system that introduced an alphabetic system to describe the sub-lexical parameters location, hand configuration and movement to give a phonological description of American Sign Language signs[34?]. A notation for a single sign consists of an initial posture description, describing non-manual features, hand-shape, hand orientation, location, and the actions changing this posture. For two-handed signs, a symmetry operator that defines how the description of the dominant hand copies to the non-dominant hand, unless otherwise specified, precedes the initial posture notation. Specifications of non-manual features and actions are optional. If the location specification is missing, a default location is assumed. In 2002, Fracik and Fabian implemented a system to visualize animated sign language sentences in real time, written in OpenGL. Their solution was part of a system to translate texts into sign language, and the animation and graphical techniques were based on a gestographic notation named easy-to-use Szczepankowski. According to their article, the system is capable to demonstrate gestures as well as whole sentences on the basis of the notation, which is easy for humans to create and read. The animation unit of this system can generate animations in real time[11]. This same year, SignStream was available as a database tool for analysis of linguistic data captured on video. “SignStream” is being designed specifically for working with data from American Sign Language, and it can be suitable to study other signed languages as well as gestural component of spoken languages. This tool consists of a collection of utterances, each one associated to a segment of video with a detailed transcription of that video. This tool also provides a single environment for manipulating digital video and linking specific frame sequences to simultaneously occurring linguistic events encoded in a fine-grained multi-level transcription. According to this article, items from different fields are vertically aligned to reflect their temporal relations. SignStream simplifies the transcription process and increase the accuracy of transcriptions (by virtue of the link of linguistic events with video frames), and it enhances the researcher’s ability to perform linguistic analyses of various kinds. In addition, multiple utterances can be open at the same time, permitting side-by-side comparison of data. There are 55 symbols

covering all the parameters[34?].

2.3.3 Robotics

In 1977, one of the first fields that started to implement projects to reduce the communication gap between not-speaking signers and speakers was Robotics. The Southwest Research Institute (SWRI) in San Antonio, TX, developed a mechanical fingerspelling hand. According to this article, mechanical hands that fingerspell may offer a solution to the inability that people with speech disability have to easily converse with others. Although the original model was unable to form all letters, the models “Dexter” and “Dexter-II” were able to display close approximations that proved to be easy to learn by deaf-blind users[19]. In 1992, Gallaudet University designed and constructed two third-generation fingerspelling for clinical evaluation named “ASEL.” ASEL was tested and configured to fingerspell hands for use with TDDs, and its letters optimization improved its recognition. As a result, a fourth-generation fingerspelling hand named RALPH was constructed by RR&D to serve as a basis for technology transfer and commercialization. Unlike its predecessors, any device with a RS232 serial data could be used to control RALPH[34].

2.3.4 Glove Based Methods

The glove methodology is one of the most studied and implemented throughout the evolution of sign language automatic interpreters. The use of gloves allows computers to track the movements of a person to express it in a spoken or written language through software. By 1992, researchers had developed cameras able to focus on people’s hand with the use of gloves with markings on the tip of each finger. In 1993, the VPL Data Glove, already mentioned in the Neural Network section, was implemented. This system has sensors that are fiber optic transducers to measure the finger flex angles. In 2002, Ryan Patterson developed a simplistic glove in the United States. The glove, a basic left-handed golf glove is outfitted with sensors and a little transmitter about the size of a 9-volt battery. A small hand-held receiver reads the hand gestures and automatically displays single lines of text[34]. Similar to a voice recognition device, the glove must be trained but in a quicker manner. The finger spelling simply spells out English words, which is equivalent to writing notes, and can only be used by Patterson’s glove. In 2003, a system for translating visual sign to written text was demonstrated. The researchers in charge of this demonstration based their system around a broad description of scene activity that naturally generalizes, reduces training requirements and allows the knowledge base to be explicitly stated. According to this article, their system description base was in a manner similar to that used in sign dictionaries. Their recognition process is performed using Markov chains to explain the temporal sequence of events at a word level, which unlike HMMs can be built from a little as a single training example[34]. In 2004, other researchers were experimenting with “mechanical skeletons”, such as that mentioned in the “Robot” section from 1994, where sensors can be placed directly on signer’s joints used in combination with an AcceleGlove, by Hernandez-Ribollar, Kyriakopoulos, and Lindeman. This same year, Hernandez-Ribollar, Kyriakopoulos, and Lindeman proposed a novel approach for capturing and translating isolated gestures of American Sign Language into spoken and written words. Their system breaks down gestures of ASL into unique sequences of phonemes, called Poses of Movements, based on software modules. According to their article, their system comprises of two main elements such AcceleGlove and a two-link arm skeleton. They improved robustness without losing portability by mounting sensors and wires of the AcceleGlove on a leather glove. Combining headgear, gloves, and a body vest provides more detailed data

on the body movements and facial expressions associated with a sign and therefore improve sign recognition. According to Hernandez-Rebollar et al., even though virtual reality gloves and suits encumber the user they are currently more reliable and efficient than using cameras[16]. By the year of 2006, the “Cyber-Glove” became available. It has a low-profile lightweight glove with flexible sensors, which accurately repeatedly measures the position and movement of the fingers and wrist. According to this article, the “Cyber-Glove” incorporates the latest high-precision joint-sensing technology, known as state-of-the-art in instrumented gloves. The resistive bend-sensing technology of this system is linear and robust, and since the sensors exhibit low sensitivity to their positioning over finger joints, and to the joint’s radii of curvature, each Cyber-Glove provides high quality measurements[30].

2.3.5 Neural Networks - Glove Based Method

Very soon robotics and cameras started to become an area of interest to researchers. Then, “Gloves” and “Cameras” were considered as analogous to a debate over using either a mouse or a keyboard as an input device. However, it was only the software that made sense of the movements. Neural Networks became an attractive technique to researchers because of the fact that they learn by example, instead of following a traditional algorithm. Researchers realized that Neural Networks could be used to extract patterns to detect complex trends because of their ability to derive meaning from a complicated data[34].

In 1993, Fels and Hinton illustrated the potential of multilayer neural networks for interfaces adapted to an individual user. They demonstrated that neural networks could be used to develop the complex mappings required in a high bandwidth interface. They used a VPL Data-Glove connected to a DECtalk speech synthesizer through five neural networks to implement a hand-gesture to speech system. According to their article, the complex mapping of hand movements to speech was learned using data obtained from a single “speaker” in a simple training phase by using minor variations of the standard back-position learning procedure. They improved their system by using separate small networks for each naturally defined subtask. Their system consisted of three Unit processes. One handles the Data-glove communication and preprocessing, another runs the stroke network, and the last runs the remaining neural networks[9]. In 1999, Weissmann and Salomon explored the use of hand gestures as a means of human-computer interactions for virtual reality applications. They published, “Gesture recognition for virtual reality applications using data gloves and neural networks”. According to their article, they explored a data glove as an input device, providing 18 measurements values for angles of different finger joints. In this article they compared the performance of the back-propagation and radial-basis neural network models functions, which were used by their recognition system[46].

2.3.6 Hidden Markov Models - Glove Based Method

Hidden Markov Models, just like Neural Networks, also became an attractive technique around the same time. In 1995, a Markov Models Chain technique was implemented on a glove based method. Kadous used instrumented gloves equipped with sensors for detecting finger bend, hand position and orientation to allow a more natural interface to computers and sign language recognition. The Glove-based Recognition of Australian using Simple Processing (GRASP) system was developed by using a simple PowerGlove from a Nintendo game unit as the capture device along with a combination of instance based and decision-tree learning to recognize isolated signs[21]. Even though this was not a hidden implementation of the Markov Model, implementation like this led to a better implementation of the Markov Model. By 1996 Lee and Xu had developed a gesture recognition system based on Hidden Markov Models. This system can interactively recognize gestures and learn new gestures online. The system is able to update its model iteratively, and it is interfaced to a Cyberglove for use in recognition of gestures from the sign language alphabet[23]. In a study by Liang and Ouhyoung in 1998, a large vocabulary was presented with a real-time continuous gesture recognition of sign language using a DataGlove™. This system first solved the end-point detention issue to statistically analyze their data based on four parameters, such as posture, position, orientation, and motion. According to their article, their system was implemented with a lexicon of 250 vocabularies in Taiwanese Sign Language (TWL), using Hidden Markov Models for 51 fundamental postures, 6 orientations, and 8 motion primitives[34].

2.3.7 Vision Based Method

In 1994, Starner wrote an article where he tracked the hands using a single camera in real-time. This was done without the aid of gloves or markings, only by using the natural color of the hands. Using vision-based sign recognition, there were two positions for the camera, the observer of the signer and the point of the view of the signer himself. In this article, Starner describes two extensible systems that use one color camera to track bear hands in real time and interpret American Sign Language using Hidden Markov Models [38][34]. In 2003, Vassilia and Konstantinos used image-processing techniques to extract useful information from signs performed and published, "Towards an assistive tool for Greek Sign Language communication." According to their article, they used Hidden Markov Models along with a language model to improve their system accuracy [45].

Vision Based Method with Accelerometer Network

This same year, Brashear, Starner, Lukowics and Junker created a lab-based Sign Language recognition system with the goal of making it a mobile assistive technology. The recognition accuracy of the system is improved by using multiple sensors for disambiguation of noisy data. According to their article, the system could be partially concealed by embedding the camera in a normal hat, such as a baseball cap, and combining visual markers and accelerometers into a watch or bracelet[3].

After Robotics, multiple projects started to be implemented in the computer vision field. Through some of these projects, many sophisticated computer graphics systems were originated leaving a broad path for research and improvement. In 2002, several parallel studies started working to improve the percentage of successfully recognized signs. This study was based on developing a vision-based sign language recognition system for mobile use. According to this article, their system uses a modified generic skin color model combined with pixel level motion information, which is obtained from motion history images [1][10][25][43]. In 2005, Gleen, Mandloi, Sarella and Lonon

created the “Sign2” project. Their project research is based on a three goals, such as body and physical movement and position enhancement to language translation, engineer a prototype device that closes the communication gap between the deaf and the hearing, and to build a statistical database from their result prototype useful to the research and development community. According to their article, their image processing techniques can be integrated with standing and developing technologies such as PDAs, smart-phones, video-phones, high-tech kiosks, among others[13]. This same year, an educational system was designed to facilitate ASL development in deaf children published in the article, “User-centered development of a gesture-based American Sign Language game”. According to this article, the system consists of three parts, such as ASL gesture recognition, a game-based interface, and an evaluation system[24].

2.3.8 Graphics-3D Animation

Avatars such as synthesized signers, virtual humans, or personal digital signers are virtual 3D animated images that can sign full words and sentences. Virtual Reality Modeling Language (VRML) is a standard language for describing 3D computer graphics models on the World-Wide Web. In VRML, 3D models provide more effective visual documentation of sign language than traditional 2D media, such as drawings and video. Therefore, the construction of ASL signs in VRML will not only help facilitate the learning of ASL by students, but also provides universal access to the learning material[34].

In 1998, there was an early project developed by Augustine Sue, which renders sign language in the form of 3D animations. This project used Virtual Reality Modeling Language (VRML) to render ASL finger spelling on the Web. VRML is an effective tool to document sign language on the World-Wide Web. According to this article, Augustine used techniques to enlarge the vocabulary of encoded ASL signs in VRML 2.0 for educational purposes. A website is used to present and demonstrate application of the hand model to fingerspell the ASL manual alphabet and numbers[40]. In 2002, a group of researcher developed a software name TESSA. This software was an experimental system that aimed to aid transactions between a deaf person and a clerk in a Post Office by translating the clerk’s speech into sign language. The system would recognize the speech and synthesize this by the appropriate sequence of signs in British sign language through a specially developed avatar. According to this article, the system software enables the communication between the speech recognition module and the avatar module to control the overall progress of a transaction. The assembly system is written in TCL and the recognition model was incorporated as TCL extension. The avatar module used by the system was written in C++, which communicated with the other system components via TCP/IP[6]. By 2002, Vcom3D (n.d.) started to develop software that incorporates avatars into educational software lessons for deaf children. Avatars can take the form of different human ethnic types or even an animal. Unlike video, avatars can be manipulated in terms of signing speed and angle. They require much less computing space than traditional video and can automatically transliterate text into Signed English, or another system. Vcom3D (n.d.) reported a research study at the Florida School for the Deaf and Blind in which comprehension of a story increased from 17% to 67% after seeing it signed (vs. being read). Seamless Solutions (n.d.) also reported that none of the students in a study they conducted had difficulty understanding the avatars, including an avatar frog. According to this article there was a study that asked two groups of non-signers to watch a sign language lesson—with either a digital video or a digital avatar. “It was found that participants learning sign language with the Avatar had a higher learning rate than video”[34][32].

In 2003, the “Auslan Tuition System” used a computer-generated signer to provide a flexible, visual Australian Sign Language (Auslan) educational tool. According to the article, the signs were generated and animated by the genetic human Modeling System that was implemented in cross-platform object-oriented C++. The Human Modeling System uses a parametric human animation technique to display 3D graphics of human upper body motion. This system was primarily designed to display sign language, but it can be used for further research into human modeling, which is the core component of the system. The system provides for input and output of XML definitions describing the hierarchical object being modeled, modifications of model joints within the subsequently generated internal kinematic tree, and forward kinematic calculations to determine global joint positions and rotations. In this project, the user has a complete viewing control of the sign display and animation, running on modest hardware. According to this article, the Rendering Module displays the position of joints of the kinematic tree graphically, while the Model Interpolation Module provides for input, output and high-level control of animations between a set of model key-frames. The system was implemented in platform-independent C++, being source-code compatible on both Linux/Unix and WIN32 architecture[34][50].

The Simon avatar system transliterates television captions into sign-supported English. It is important to know that closed captions are not as effective for a deaf person as subtitles in a foreign movie are for a hearing person since it is not deaf the person’s native language. The first benefit is the ability of the consumer to choose the avatar as opposed to scrolling text. A second benefit is that the avatar is only visible when activated to avoid disturbing hearing viewers. The software is customizable and less expensive than a live human. Model signs for Simon were acquired from expert signers using a Cyber-Glove, an Ascension Motion Star wireless magnetic body suit, and an optical Facetrak device, for facial expression and lip position. When a word needs to be transliterated, it is found in the dictionary and matched to the accompanying physical movement, facial expression, and body positions, which are stored as motion-capture data. The ultimate goal is to add an NLP element so that BSL can be shown instead of sign-supported English[34].

2.4 Current System Implementations

The ASL automatic interpreter system is in its relatively early stages of development. At the present time, a computer program can convert Sign Language into text and a computer-generated voice. However, the speed at which sign to speech and text conversion is proceeding appears more rapid because at the present time there is considerable momentum on an international scale behind this technology. Multiple projects, similar in design and approach, are presenting opportunities for information exchange. These projects are underway in a number of jurisdictions, leading to an international high level of interest in this evolving technology with the development of projects, such as the Atlas program in Italy, the DePaul ASL Synthesizer Project in the U.S., SASL-MT in South Africa, and the ‘SiSi’ (Say It, Sign It) project in the U.K. These projects typically use linguistic processing. This means, teletext analysis and speech recognition software to create sequences of motion-captured signing data. More recently, this approach has been enhanced with synthesized animation from HamNoSys (the internationally established phonetic transcription system for Sign languages), which is integrated with an avatar animation platform. This is necessary because HamNoSys does not transcribe facial expressions. The avatar platform combines skeletal animation with accurate facial gestures. This is becoming increasingly sophisticated in its ability to record and animate more complex gestures, body language and facial expressions[4].

Until around 2009, there were some applications that were able to convert speech or text to sign. These applications only matched individual words to individual signs out of a sign language dictionary. Their speed and accuracy made them a communication tool, but not an interpreter substitute. The technology focus has shifted from telecommunications to communication applications, where software applications are used to deliver a new type of communication between users. After 2009, research and development in the field of assistive technologies like speech to text and sign to text/speech conversion gave a rise to another, potentially major spin-off: a more universal translator that automatically translate text from one sign language to a spoken language, or translates the spoken words of a language to a sign language[4]. This same year, the Sign language converter necklace or pendant was announced. The device, essentially a sign to speech translator, can be worn around the neck to pick up sign language and convert the symbols into speech[4]. Another, “Sign Voice Language Translator”, also worn as a pendant around the neck, uses a camera to capture and translate Sign language into speech, and then converts the speech into text that appears on a small LCD screen next to the camera. This would enable an individual who is blind and, or, deaf to communicate with one another[4].

In 2010, the SignSpeak project was said to be the first step to approach sign language recognition and translation at a scientific level already reached in projects such as automatic speech recognition or statistical machine translation of spoken languages. The goal of this project was to develop a new vision-based technology for recognizing and translating continuous sign language to text. According to this article, the new technology of sign language structure from the perspective of machine recognition of continuous sign language will allow a subsequent breakthrough in the development of a new vision-based technology for continuous sign language recognition and translation. The SignSpeak project used innovative scientific theory and vision-based technology development by gathering novel linguistic research and the most advanced techniques in image analysis, automatic speech recognition (ASR) and statistical machine translation (SMT) within a common framework[8]. In 2011, IBM combined a number of computer technologies including speech recognition, which converts spoken word into British Sign Language. This is then signed by an animated digital character or avatar which pops up in the corner of a display screen (computer,

laptop, television screen). The Open Sign database, an international compilation of Sign language projects and research currently underway, indicates that nine such sign and avatar initiatives are currently underway in Europe and South Africa[4].

In 2012, a slight variation of the above automatic interpreter programs is Dicta-Sign, a project in development at the Athena Institute of Language and Speech Processing in Greece. The project enables Sign Language interaction with Web 2.0, so that a person who ‘dictates’ changes via sign language. These signs are then converted to an avatar who signs them back to users, who can make updates and contributions[4]. In the same year, a group of Ukrainian students organized in a venture called “Enable Talk” has developed a set of gloves that, when connected to a smart-phone via Bluetooth, automatically translate Sign language to text, and then into speech. The gloves use “flex sensors, touch sensors, gyroscopes and accelerometers (as well as solar cells to increase battery life)” to produce a relatively inexpensive system that can adapt to a range of international Sign languages[4]. In 2013, an automatic sign language to speech and text interpreter started a new project at a firm named “Technabling.” Technabling is a spin-off company of the University of Aberdeen in Scotland. The interpreter is named Portable Sign Language Translator (PSLT). PSLT generated attention due to its groundbreaking advancements in translating signs to speech and text[4].

In 2013, the Sign Language Ring was one of the winners of the 2013 Red Dot Design Award. A group of researchers from Asia University invented the Ring concept. Their system is composed of two simple wearable devices, such as the bracelet and the rings. Six rings are worn on the thumb, pointer and middle finger of each hand. The rings are digitally enhanced in order to understand and interpret the movements of the hands as they relate to sign language. Then, these rings communicate the words to the bracelet, which emits a voice translation to nearby listeners. The rings and the bracelet can be programmed to understand customized movements for easier communication. Also, the system can work in reverse order by picking up words of other people and convert them into digital text, which reads out on the bracelet. In this way, people with hearing impairment can communicate easily with anyone[31].

2.5 Limitations in Current Systems Implementations

Sign Language is a type of communication that uses gestures as well as facial expressions and body language to convey meaning. Current Sign-to speech and text conversion technology cannot capture all of these sometimes-subtle nuances. Next, syntax is also very important; that is, Sign Language is not directly translatable word-for-word to spoken language and some degree of interpretation is typically required to understand context and meaning. Others central barriers are:

- The extent to which the software can read and convert specific signs
- Accuracy of voice recognition software; it may be used to convert Sign Language to speech
- Speed at which the software and camera can capture signs

The limitations of voice recognition software noted above also apply to sign to speech conversion.

DEXTER

- The mechanical hand could not exactly mimic the human hand in fingerspelling all the letters (such as the special wrist and arm motions required in J and Z. The inter-letter neutral position was another unnatural feature of the design that did not accurately reflect 240 Journal of Rehabilitation Research and Development Vol . 31 No. 3 1994 human fingerspelling and limited the speed of letter presentation.
- A modular hybrid design approach allows the best-suited algorithm to be applied at each conceptual stage of the processing, and provides for their integration at each development stage. This modular approach begins with a limited vocabulary and a restricted environment.
- Because the hand shape will require the highest level of resolution, and is statistically the least accurate part of sign production, the early development stages will de-emphasize the use of hand shape for feature extraction. Later stages will need to be invariant to hand shape ambiguities and mistakes.

The Auslan Tuition System

- The Auslan Tuition System is able to effectively demonstrate signs with these attributes. However, at this stage of development, facial expressions cannot be displayed. Current research has begun to address the addition of facial expression to the system.

Kinect

- The hierarchical CRF is useful for recognizing hand motions; however, it has difficulty distinguishing between different hand shapes.

Leap Motion

- For the static recognition problem each detected hand's position and orientation are being treated as a new data point to easily generated multiple samples from the sensors in short

time, and treat this as a classification problem. The 3D data yield by the sensor is represented in the sensor's coordinate system. Therefore, even a small change of the position or orientation of a hand with respect to the location of the sensor can negatively affect the gesture recognition system.

- The proposed approach can be treated as state-of-the-art approach, but it still cannot be used without defining proper feature sets for gesture recognition.
- Another problem encountered using this approach is the fact, that matrices no longer represent the probability, as the total values in rows do not add to 1:0. Therefore, each row of new matrices is normalized to 1:0. When dealing with multiple training examples, the learnt solution may adapt to the training samples, but perform poorly on the testing set.

2.6 Promising Graphics System Implementation

Automatic interpreter technology is a software-based system that converts Sign Language to computer-generated spoken or written words, or converts text or spoken word to Sign Language in “real time”[4]. Computer graphics is the most suitable technology to implement ASL on an automatic translator. It supports immediate creation of new animations based on the existing rules and conditions that changes input from outside sources, and provide the flexibility necessary for ASL sign translation. This is why applications in sign to speech and text translation utilize existing motion capture technology to create a signing avatar, which it is also used in video gaming system such as Wii, Xbox One Kinect, and Leap Motion Controller. Most of the current, and promising, advancement in ASL automatic translator has been obtained through devices, as those mentioned above, that provide you with accurate 3D data about users movements.

In 2014, a group of researchers developed a project named “Hand in Hand: Automatic Sign Language to English Translation.” In this article, they described their system as the first data-driven automatic sign-language-to-speech translation. According to their article, they combined a sign language recognizing framework with a state-of-art phrase-based machine translation system, using corpora of both American Sign Language and Irish Sign Language data. Their system was based on a automatic speech recognition system adapted to visual features. The word sequence, which best fits the current observation to the trained word model inventory and language model will be the recognition results. Their baseline system was Viterbi trained, which uses a language model[39].

In 2014, Matt Huenerfauth published “Learning to Generate Understandable Animations of American Sign Language.” In this project, Huenerfauth express how his lab, stationed at Rochester Institute of Technology, has designed new algorithms to select movement details for ASL animations, given, a script with sparse information about the sentence. According to this article, the lab has identified speed and timing for ASL animations, placement of pauses during sentences, planning of complex sentences called “classifier predicates”, temporal coordination during animations, and the linguistically accurate selection of the motion-path and hand orientation of ASL verbs. These indicate locations in 3D space around the signer where entities under discussion have been established, which has been data-driven to make use of statistical models trained from examples of human movement collected in his ASL motion-capture corpus. Huenerfauth team is currently designing models that link the linguistic phenomena with specific facial movements to produce animations. They are currently building an infrastructure for synthesizing animations using MPEG-4 facial animation parameters. The lab is also conducting periodic evaluations with native ASL signers to evaluate animations with facial expressions synthesized based on their computational linguistic models[18].

Kinect Sign Language Translator

In 2013, the “Kinect Sign Language Translator”, which is currently under development, was announced, and its results seem promising. Dedicated researches in China have already crated a prototype of the Kinect Sign Language Translator. This system understands the gestures of sign language and converts them to spoken and written language and vice versa. According to this article, the system can capture a conversation from both sides. For example, the signer is shown signing, with a written and spoken translation being rendered in real-time for the hearing person. The non-signer is represented by an avatar that takes his or her spoken words and turns them into accurate and understandable signs. These signs accompany the written translation for the deaf person to “read,” all in real-time. This project is a potential game-changer for many in the signing and non-signing communities alike[48].

MotionSavvy Sign Language Translator

In 2014, Ryan Hait-Campbell, CEO and founder of Alameda, California based MotionSavvy announced a project named “MotionSavvy”, which it is still under development. He, along with his colleagues who are also deaf, developed a groundbreaking technology that combines the latest in motion sensing and mobile computing called UNI. According to this article, the UNI has three parts such as the tablet computer, a specially designed smart case, and a mobile application. The smart case contains hardware from Leap Motion, and it consists of a couple of cameras to track the location of the user’s hands and fingers. The tablet powers the application and it translates the hand and finger movements of a sign language into audible speech or text displayed on the screen. The application in turn can also translate spoken word into written text for the deaf person to read. UNI does more than interpreting in this system, it also learns. In the same way spoken language, sign language consists of various “dialects” and “accents”. UNI learns your own style of signing as you train it to improve its accuracy. It is expected to grow smarter through a database of new gestures that are “crowd-signed” by its users[20].

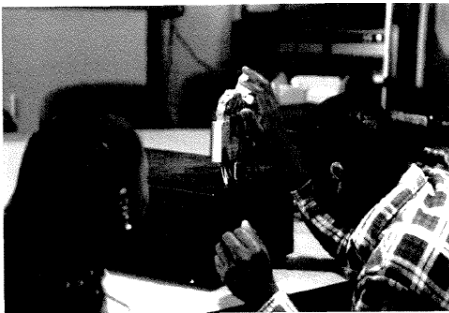
3 ASLT Attractive Implementations

3.1 Glove Based Methods

3.1.1 Neural Netowrks

Neural networks are used to extract relevant features of sign language from video images of a person communicating in a Sign Language. The key features are hand motion, hand location with respect to the body, and hand shape. In 1985, the Rehabilitation Engineering Center of The Smith-Kettlewell Eye Research Foundation sponsored a class project. The major goal was to develop a system with improved timing and easily modifiable finger positions. These qualities were realized in a new robotic fingerspelling hand named “Dexter” shown in the image below. Signal processing techniques developed for defense-related programs were adapted to this project. Algorithm development and transition using neural network architectures were encouraged[19].

Figure 1: Leap Motion Functionalities



Implementation Approach:

The microcomputer and its associated software controlled the opening and closing of the bank of valves, which directed air pressure to specific pneumatic cylinders which pulled on the drive cables and were the “tendons” of the fingers. As a message was typed on a keyboard, each letter’s ASCII value was used by the software as a pointer into an array of stored valve control values. The states open or closed of all 22 valves were specified by three bytes. Two to six valve operations, each separated by a programmed pause, were needed to specify the finger movements corresponding to a single letter. The hand could produce approximately two letters per second, each starting from and returning to a partially flexed neutral

position. The software to determine whether the current finger position was an intermediate or final letter position used an additional bit in the valve control byte triad. Different programmed pause times were associated with each of these two situations. The mechanical hand could not exactly mimic the human hand in fingerspelling all the letters, such as the special wrist and arm motions required in J and Z. The inter-letter neutral position was another unnatural feature of the design that did not accurately reflect 240 Journal of Rehabilitation Research and Development Vol . 31 No. 3 1994 human fingerspelling and limited the speed of letter presentation[19].

Overall Approach

Features must be extracted for many users with a variety of hand sizes, who each use different signing styles, speed, and clarity. The approach currently being developed involves the interaction of many advanced techniques. In addition to neural network algorithms, tile design involves the integration of numerous image processing techniques, infrared and CCD sensor technology, wavelet analysis, fuzzy logic, and expert systems. A modular hybrid design approach allows the best-suited algorithm to be applied at each conceptual stage of the processing, and provides for their integration at each development stage. This modular approach begins with a limited vocabulary and a restricted environment. The first stage of development provides location, and movement labels, and coarse hand shape descriptions. The beginning features have been chosen to allow early

implementation of a limited unit that has the largest vocabulary possible for such a stage. The translation system requires the use of at least one camera to capture the images of the signer. Their development work has been with CCD and infrared cameras. Sensor fusion techniques were applied to allow for the integration of both camera types. The second stage of the system converts the running video into sequences of digitized frames. The video must be processed in real time, so compression techniques are employed. Some of the algorithms require less resolution, so it is not necessary to present every frame to every part of the process. Capturing high-resolution frames at a high frame rate, and dropping a number of frames in between processes to reduce the effective rate, can achieve real time computation[19].

The computer implements a number of image processing and signal processing algorithms to parse the image data into sign parameters. The algorithms interact with each other at various levels of complexity to produce a sign parameter triplet of location, movement, and hand shape. The coarsest level of processing identifies the hand and its relation to the body. The “head” and “body” are identified as locations first. The next level of location labels builds on the “head” description to include the difference between “forehead” and “chin,” and builds on the “body” to provide “shoulders” and “waist” as labels. People use their sign space differently and produce signs differently, so location labels will not be exact. Their design goal is to produce a decision on the parameter triplet and assign probabilities values to the parameters for alternatives to be readily produced when the most likely does not make sense. The information about the hand placement in a single image used to correlate the hand with its location label will also be used to define movement. Successive hand placements will be used for the movement algorithm. Early development stages will detect and track simple movements such as “up,” “down,” “in,” “out,” and “side-to-side.” Later stages will identify “rotation” and “shake” movements. Comparison between sequential frames will provide much of the information needed for the movement classification. Also, it will label the dominant hand as opposed to the non-dominant hand and the face. In standard or infrared video images, the hands and face exhibit the same contrast to the background. As a result, the first level of image processing will identify both hands and the face as the potential dominant hand to track. The hand placement information will also be used to segment the hand portion of the image for use by the hand shape classification algorithm. At the coarsest level, the hand shapes will be grouped into similar shapes. Because the hand shape will require the highest level of resolution, and is statistically the least accurate part of sign production, the early development stages will de-emphasize the use of hand shape for feature extraction. Later stages will need to be invariant to hand shape ambiguities and mistakes[19].

3.1.2 Hidden Markov Models

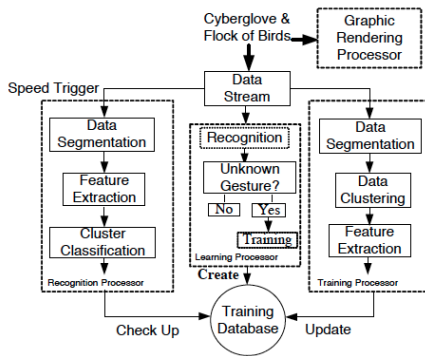
A CybergloveTM sensory glove and a Flock of Birds[®] motion tracker are used to extract the features of ASL gestures. The data obtained from the strain gages in the glove defines the hand shape while the data from the motion tracker describes the trajectory of hand movement. The objective of this project is to continuously recognize ASL gestures using these input devices in real time. With the features extracted from the sensory data, they specify multi-dimensional states for ASL signs in the Hidden Markov Models (HMM) processor. The developed system forms a sound foundation for continuous recognition of ASL full signs[23].

Hidden Markov models possess the ability to segment the data to its constituent signs implicitly and continuously. To achieve continuous recognition of ASL signs they developed a stochastic process on the HMM algorithm. The data from the “Cyberglove” and Flock of Birds sensors is clustered to extract the gesture features using the LBG algorithm. The parameters of the HMMs are defined with Gaussian distributions based on the training samples[23].

Overview

Hidden Markov Models are a type of doubly stochastic models. In developing their system they used the discrete HMM, where the observations are characterized as a finite set of symbols. A typical discrete HMM can be specified by N distinct states, M distinct observation symbols per state, and the probability matrices of state transitions, observation symbols, and the initial state of the HMM process. A discrete HMM can be presented as $\lambda = (A, B, \pi)$, where, $A = \{a_{ij}\}$ is a matrix of the state transition probability distribution, and a_{ij} specifies the probability that state S_i changes to state S_j ($1 \leq i, j \leq N$); $B = \{b_j(k)\}$ is the observation symbol probability distribution in state S_j , and it represents the probability that the system will output an observable symbol O_k at the state S_j ($1 \leq j \leq N; 1 \leq k \leq M$); π is a vector representing the probability that each state is the initial state of the HMM process[23].

Figure 2: Cyberglove HMM Design



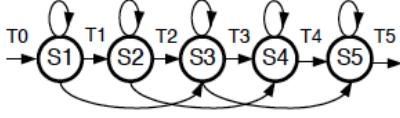
There are three fundamental tasks in the HMM design, Figure 2: (1) Given an observation sequence, compute the probability with which those observations generated by a given HMM model; (2) Determine the most likely sequence of internal states in a given model which will give rise to a given observation sequence; (3) Adjust the model parameters of a HMM to optimize the probability distribution matrices for a given set of observations[23].

Multi-dimensional HMM

In this model, each dimension of the HMM state corresponds to the data from each sensor channel. The multiple data streams from the sensory glove and 3-D motion tracker are the inputs to the HMM process, and this raw data corresponds to the sequence of observations in the HMM. There are 21 channels of raw data; one 21-dimensional state was defined for each ASL alphabet in the HMM model. The probability matrices of the HMM (A, B, π) are specified by clustering Gaussian distribution. A 5-Bakis-state

HMM was used in the system, as shown in Figure 3, one state in an HMM can reach the same state or one of next two states[23].

Figure 3: 5-Bakis-State HMM



In the alphabet recognition stage, these successive readings are typically classified as the same state if the user intends to gesture a certain ASL alphabet. Due to the large amount of words spelled by alphabets, equal probabilities for the transition of internal states and the initial state of a HMM process was assumed. Clustering the Gaussian distribution defined multiple B matrices for the dimensions of each HMM state. Since a discrete HMM

was being used, a gesture was represented as a sequence of discrete symbols. The raw gesture data was then process, which in this case were the values of the 15 bending and abduction angles, 3 position coordinates and 3 orientation angles, which were obtained from the “Cyberglove” and the Flock of Birds tracker. The retrieval rate of the raw data was 40Hz, and the system segments the data stream based on a velocity trigger. The segmentation procedure started when the hand is stationary, i.e. the speed of the hand of the user was below a preset threshold, and it ended when the speed exceeded the threshold[23].

Probability Function Calculation

A continuous Gaussian probability density function with mean μ and standard deviation σ can be defined as[23].:

$$P(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

$$\int_{-\infty}^x P(t)dt = \int_{-\infty}^0 \left(\frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{t^2}{2}} \right) dT + \int_0^x \left(\frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{t^2}{2}} \right) dT = \frac{1}{2} \int_{-\infty}^0 \left(1 + erf\left(\frac{x}{\sqrt{2}}\right) \right) dT$$

$$f(x) = \int_{-\infty}^x P(u)du = 1 - P(x)[C_1Xt + C_2xt^2 + C_3xt^3] + \epsilon(x)$$

$$erf(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-t^2} dt$$

where $P(u)$ is the standard Gaussian probability function,

$$t = \frac{1}{1+0.33267Xx}, C_1 = 0.4361836, C_2 = -0.1201676, C_3 = 0.9372980, \text{ and } |\epsilon(x)| < 1 \times 10^{-5}$$

Where $x \in (-\infty, +\infty)$, μ is the clustered feature of a set of N observations ($X_1, X_2, \dots, X_N$), and σ is the distortion over the N samples. The cumulative Gaussian distribution function can be obtained with the following equation, where erf is the error function[23].

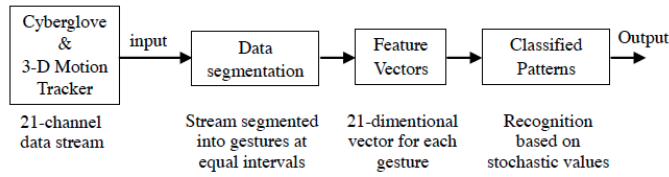
Gesture Recognition

The recognition process dealing with the data acquisition, segmentation, feature extraction, and pattern recognition is implemented as follows. Having a sequence of observations, i.e. the raw data stream from the sensors as an input to the HMM process. The most likely sequence of internal states needs to be determined, $S_{opt} = \{S_{opt1}, S_{opt2}, \dots, S_{optL}\}$ for the given observation sequence $X = \{X_1, X_2, \dots, X_L\}$ such that the probability $P[S_{opt}, X | \lambda] = \max(P[S, X | \lambda])$, where S is a random sequence of L internal states. Because of the discrete HMM model, the state sequence S_{opt} can be found by calculating the maximum probability, $\max(P[S, X_i | \lambda])$, where $S_{arb} \in \{S_1, S_2, \dots, S_N\}$ and $X_i = \{X_1, X_2, \dots, X_L\}$. For each of the internal states the proba-

bility matrices B is created by inputting the sequence of observations (21-dimensional vectors of sensor readings) to the HMM system. This calculates the integral of Gaussian function for those inputs based on the clustered codebook. The various readings of each sensor are then quantized as a single value that represents $b_j(k)$. Therefore, there are 21 observation probability matrices of B [23].

ASL Recognition

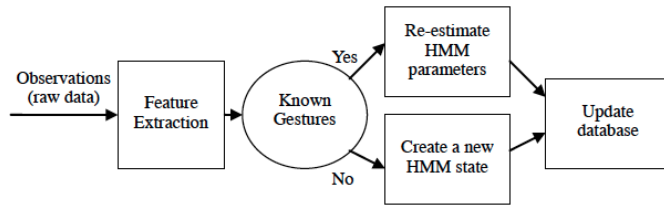
Figure 4: Recognition System



gram will start to segment the data stream to extract the features of the current gesture. The gesture feature then is ready for further processing. Fig. 5 illustrates the data flow in the HMM recognition processor[23].

Interactive Training/Learning

Figure 5: Data Flow



record-oriented files used to store each HMM state as a record with a corresponding index. One record in the database generally includes two 21-dimensional vectors of mean values and standard deviations[23].

3.1.3 Vision Based Methods

Proposed Solution

The system concentrates on the practice and correct repetition of ASL phrases and allows the child to communicate with the computer via ASL. The project targets children at the Atlanta Area School for the Deaf (AASD), ages 6-8, who are behind in their language development[24].

Figure 6: Game Interface



a) Tutor Video b) Live Camera Feed c) Attention Button d) Animated Character and Environment e) Action Buttons

The game involves Iris the cat who can interact with her environment in a number of ways based on what the child signs to her. With the assistance of educators at AASD, a list of eight age-appropriate ASL phrases was developed[24].

At first, Iris a cat is asleep. The child can press an Action Button (Fig. 6e) and watch a short video clip showing the correct ASL command in the Tutor Window (Fig. 6a). Next, the child clicks on the Attention Button (Fig. 6c), and Iris wakes up. The child then signs the command to Iris. A live video feed allows children to see themselves (Fig. 6b). If the phrase is correct, Iris carries out the command or answers the child's question with a pictorial thought bubble in the window (Fig. 6d). If the child does not sign correctly, a thought bubble with a question mark appears above Iris's head. Eventually, a gesture recognition system will control the computer's response to the child's signing. However, while the gesture recognition system is under development, the initial game design is initiated via Wizard of Oz (WOz) methods. The initial goal was to design a prototype for the game that the children would find engaging and interesting enough to practice their signing. By observing how the children use the system, they enhanced usability and enjoyment. They decided to use children slightly older than the targeted age (ages 9-11) for short pilot studies because they were capable of giving more feedback about the interface. Student in groups of 2 or 3 were tested, making subtle changes based on the previous groups' opinions and feedback. This diverse sample set allows for testing and development of their gesture recognition software with a variety of signers, signing techniques, skin tones, and physical attributes. This data set is invaluable for training of the mathematical models and performance testing of the gesture recognition system[24].

3.1.4 Graphics - 3D Animation

There are several components of sign language that must be effectively communicated in order for a learner to understand and replicate a previewed sign. Signs comprise hand shape, location, orientation, movement, and facial expression. The most important is the shape made by the arms and hands. Sign hand shapes are applied to hands depending on whether signs are one handed, double handed or two handed. In one-handed signs, only the primary hand moves and has an associated hand shape. Double-handed signs use the same hand shape on both hands, whereas two-handed signs have two different hand shapes. There are 38 major hand shapes in Auslan, with a further 28 variants. Location of the hands is an important feature and includes positions of contact between the hands and body, and between the two hands, while orientation refers to the direction in which the palms and hands point. These shape attributes of a sign must be clearly visible to the user. Movement is also a key aspect of sign language and they must appear natural and be clear enough to be understood by the user. The Auslan Tuition System is able to effectively demonstrate signs with these attributes. However, at this stage of development, facial expressions cannot be displayed. Current research has begun to address the addition of facial expression to the system[50].

Figure 7: System Interface



face also highlights the current word being signed, and provides links to detailed information about the individual signs. Figure 8 (a) shows a sign being animated from a sign list[50].

Figure 8: Sample System (a)

User Control and Interactive Features

Position and movement are necessary to correctly produce signs. Familiar “VCR-style” playback controls and simulated frame-by-frame positioning of the animation allow specific sections of signs to be examined more closely. Figure 9 (b) shows a screenshot of a sign being animated from this first-person view. The flexible rendering and interpolation system, used to display signs, facilitate adding interactive features to enhance the learning experience of the user. Fingerspelling is used for proper nouns, where there is no translation for an English word the system is able to render an English phrase input by the user as the fingerspelling Auslan[50]. An example is shown in Figure 8 (a).

Displaying Sign Language in Context

Similarly to spoken languages, Auslan usage changes depending on conversation context. We have created an Auslan dialogue section that allows the user to view an Auslan conversation between two rendered signers, showing the animated signs, sign names, and English meanings concurrently. This interface helps to highlight situations where signs presented in the Tutorial section would differ in the context of a conversation[50].

Modification of Tutorial Contents

To build signs for the Auslan Tuition System a Sign Editing Interface was designed. The interface allows the user to create static poses, by visually selecting joints of the model to rotate, then using Euler angle modification controls to position them. A screenshot of the sign-editing interface is shown in Figures 7, 8, and 9. The system provides pre-defined positions for the major Auslan hand shapes as a starting point for creating poses. Static poses can be saved, or sequenced into sign animations. Changing the timing of sequenced key-frames controls the speed of the animation. Signs are saved to disk as Key-frame-Sequence XML files. The Tutorial System loads and parses Sign-Category and Sign-Dialogue XML files to populate sign tutorial and sign dialogue categories and sign lists. These XML files store English descriptions, and sign control strings that set which signs to load at specific times. A simple parser reads, loads and sequences signs, loading sign key-frames directly from Key-frame-Sequence XML files stored within the working folder of the Auslan Tutorial System executable. Adding signs to the system is therefore a simple exercise — first, the sign is created using the sign editor, and saved into the correct directory on the disk, then the corresponding Sign-Category or Sign-Dialogue XML file is modified to include a phrase with the newly created sign. New categories of signs can also be added by modifying the Sign-Category XML file[50].

Targeting Educational Content

Sign language literature and systems can be roughly categorized as either educational or reference material. The sign tutorial section combines signs to form phrases, and groups these phrases into progressive categories. The inter-

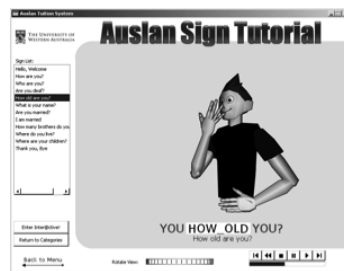
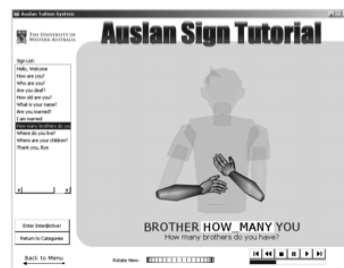


Figure 9: Sample System (b)



3.1.5 Microsoft Kinect as a Translator

In order to understand how and why Microsoft Kinect is developing an automatic ASL translator, I will briefly explained the approach used in the article “Sign Language Recognition with the Kinect Sensor Based on Conditional Random Fields” [49] to convert Kinect to a Sign Language Translator. According to this article, sign instances vary in motion and shape in three-dimensional (3D) space, which makes difficult for recognition. The method proposed in this article used 3D depth information from hand motions, generated from the Kinect devices sensor and apply a hierarchical conditional random field (CRF) to recognize hand signs from the hand motions. Then, to verify the hand shapes of the segmented signs a BoostMap embedding method is used.

Face and Hand Detection

The Kinect Windows software development kit tracking function was used to robustly detect the face and hand positions. The skeletal model consisted of 10 features from the upper body, as shown in the figure below[49].

Figure 11: Skeleton Model

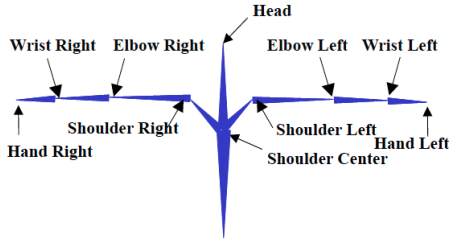
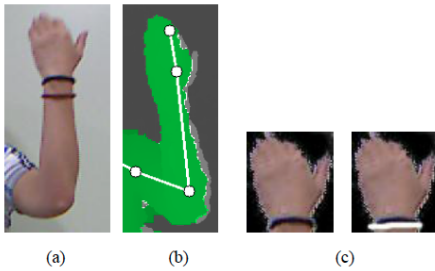


Figure 2. Skeleton model: 10 upper body components.

Figure 12: Arm and Hand Extraction



The hand region is obtained by establishing a threshold from the hand position as shown in the Figure (a). The signer wears a black wristband to segment the hand shape. Random Sample Consensus (RANSAC) is used to detect the black wristband, as shown in the Figure (c). The detected hand shape is normalized. (a) (b) (c)[49].

Hand detection: (a) color image (b) depth image and feature positions (c) and detected regions with black wristband[49].

Feature Extraction

Features are extracted in 3D and 2D space using the detected hand and face regions as shown in Table1[49].

The feature, HFL , represents the location of the left hand with respect to the signer's face in 3D space. The distance between the face and left hand, $DHFL$, and the angle from the face

Figure 10: Recognition System

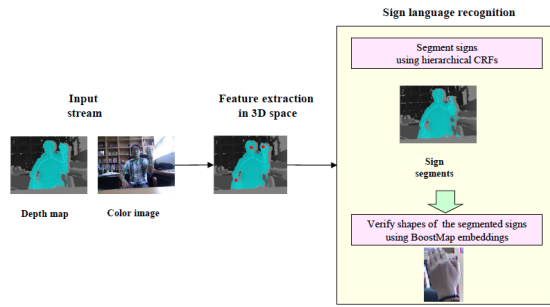


Figure 1. Overview of the proposed method for recognizing a sign.

to the left hand, θHFL , is measured. In order to extract 3D features, the coordinates of the left hand are projected into the x, y and z axes. The angle between the face and left hand, $\theta HFL = \{\theta x, \theta y, \theta z\}$, is extracted. Then, the feature vector $\{DHFL, \theta HFL\}$ is clustered into an index using an expectation-maximization (EM)-based Gaussian Mixture Model (GMM). Features, HHL, HFR, HHR, FSL and FSR , are likewise calculated and clustered[49].

Figure 13: Table of Features

Table 1. Seven features for recognizing the signer's hand.	
Features	Meanings
HF_L	Position of the left hand with respect to the signer's face
HF_R	Position of the right hand with respect to the signer's face
HH_L	Position of the left hand with respect to the previous left hand
HH_R	Position of the right hand with respect to the previous right hand
FS_L	Position of the left hand with respect to the shoulder center
FS_R	Position of the right hand with respect to the shoulder center
OH_{LR}	Occlusion of two hands

The hand occlusion, OH_{LR} , is determined from the ratio of the overlapping regions of the two hands in the frontal view:

Where HI is the left hand region, HR is the right hand region, R_O is the overlapping region between the two hands, and T_O is the threshold for hand occlusion ($T_O=0.3$, as determined by experimentation)[49].

Figure 14: Equation (1)

$$OH_{LR} = \begin{cases} 1, & \min(\frac{R_o}{H_l}, \frac{R_o}{H_r}) > T_o, \\ 0, & \text{otherwise,} \end{cases}$$

CRF-Based Sign Language Recognition

A hierarchical CRF framework recognizes the sign language. First, a threshold model ($T - CRF$) is distinguished between signs and non-sign patterns, where the label " $N - S$ " defines non-sign patterns and the labels in the vocabulary define signs. When constructing the $T - CRF$, a conventional CRF is constructed first. The conventional CRF includes the labels $SC = \{Y1, \dots, Yl\}$, where $Y1$ through Yl are labels for signs, and l is the number of signs in the vocabulary[49].

In a CRF, the probability of a label sequence y , given an observation sequence x , is found using a normalized product of potential functions. Each product functions is represented by:

Where $F_\theta(y_i - 1, y_i, x, i) = \sum_v \lambda_v t_v(y_i - 1, y_i, x, i) + \sum_m \mu_m s_m(y_i, x, i)$, $t_v(y_i - 1, y_i, x, i)$ is a transition feature function of the observation sequence x at positions i and $i - 1$, where $s_m(y_i, x, i)$ is a state feature function of observation sequence x at position i , $y_i - 1$ and y_i are the labels of observation sequence x at positions i and $i - 1$, and λ_v and μ_m are the weights of the transition and state feature functions. θ represents the weights of the transition features and state feature functions, and $Z_\theta(x)$ is the normalization factor[49].

Figure 15: Equation (2)

$$p_\theta(y|x) = \frac{1}{Z_\theta(x)} \exp(\sum_{i=1} F_\theta(y_{i-1}, y_i, x, i))$$

The feature vector x_t , of the observation sequence x , at time t , is expressed as:

$$x_t = \{HL_L^t, HR_R^t, HH_L^t, HH_R^t, FS_L^t, FS_R^t, OH_{LR}^t\}$$

CRF parameter learning is based on the principle of maximum entropy. Maximum likelihood training selects parameters that maximize the log-likelihood of the training data. The $T - CRF$ is built using weights from the constructed conventional CRF . In addition, the label " $N - S$ " for non-sign patterns is added to the conventional CRF . Thus, the $T - CRF$ includes the labels $ST = Y1, \dots, Yl, N - S$. The starting and ending points of in-vocabulary signs were calculated

by backtracking the Viterbi path, subsequent to a forward pass[49].

The weights of the transition feature functions from other labels to the non-sign pattern label “ $N - S$ ” and vice versa are assigned by:

Figure 16: Equation (3)

Where $\lambda_v(N - S, N - S) = \text{argmax}_k \lambda_v(Y_k, Y_k) + k$, and $k(k = 1, \dots, l)$ is the weight of the self-transition feature function of the non-sign pattern label “ $N - S$.” After constructing the $T - CRF$, the second layer CRF , which models common sign actions, is constructed. The output of the first layer is the input of the second layer. It contains the segmented signs, which signs have a higher probability than the non-sign pattern label “ $N - S$ ”. As a result, the second layer CRF only has labels $SC = Y1, \dots, Yl$ [49]. Finally, the probability of the recognized sign is calculated as:

$$\forall_{k \in \{1, \dots, l\}} \lambda_v(Y_k, N-S) = \frac{\lambda_v(Y_k, Y_k)}{l},$$

$$\forall_{k \in \{1, \dots, l\}} \lambda_v(N-S, Y_k) = \frac{\lambda_v(N-S, N-S)}{l},$$

Figure 17: Equation (4)

$$P(y_i^t) = \frac{\sum_{i=s_s}^{s_e} p_\theta(y_i^t | \mathbf{x})}{s_e - s_s + 1}$$

Where $p_\theta(y_i^t | x)$ is the marginal probability of the sign y_i at time t ; s_s and s_e are the start and end frames of the segmented sign, respectively[49].

Shape-Based Sign Language Verification

The hierarchical CRF is useful for recognizing hand motions; however, it has difficulty distinguishing between different hand shapes. The main goal of the hand shape-based sign verification is to determine whether or not a sign spotted through the $H - CRF$ should be accepted as a sign. The shape-based sign verification module is performed at the end frame of a recognized sign, when $P(y_i^t)$ in the Equation above is lower than a threshold[49].

BoostMap embedding is applied in order to recognize the hand shape. This method accommodates various scales, rotations, and sizes of the signer’s hands. Synthetic hand images to train the model are generated using the Poser 7 animation software. Each sign begins and ends with a specific hand shape, and each alphabet has unique hand shapes. Table 2 and Figure 4 show examples of hand shapes for sign language recognition. The system uses a database with 17 hand shapes. For each hand shape, 864 images are generated[49].

Figure 18: Eq. (5)

$$V_s(y_i^t) = \sum_{j=t-t_a}^{t+t_a} C_a(y_i^t, B(j)),$$

$$C_a(y_i^t, B(j)) = \begin{cases} 1, & y_i^t = B(j), \\ 0, & \text{otherwise,} \end{cases}$$

The hand shapes are verified over several frames, and a detected sign is accepted when the voting value $v_s(y_i^t)$ exceeds threshold T_s . The voting value, $v_s(y_i^t)$ is calculated as:

Where (y_i^t) is the sign detected by the $H - CRF$ at position t , and t_a is the window size. $C_a(v_s(y_i^t), B(j))$ is:

Where $B(j)$ is the recognition result of the BoostMap embedding

method at time j .

O stands for one-handed sign; T stands for two-handed sign; D.C. means don’t care; $>$ Means that the hand shapes of start and end frames of the sign are changed[49].

3.1.6 Leap Motion as a Translator

The technology used in the “MotionSavvy” sign language translator, just as with the Kinect translator, has not yet been disclosed to the public. However, similarly to the Kinect, many approaches for Leap Motion gestures and signs recognition have been successfully implemented. To understand the approach utilized in the MotionSavvy project, I will briefly explained the approach used in the article “GESTURE RECOGNITION LIBRARY FOR LEAP MOTION CONTROLLER.” I must emphasize that this article uses a Support Vector Machines (SVM) implementation, and a Hidden Markov Model implementations. These techniques have been a huge influence in the automatic translator system evolution[29].

Leap Motion Approach - Implementation

The gestures can be divided into two main groups: statics gestures and dynamic gestures. The static gestures can be understood as a position and orientation of fingers and hand in a single moment, where SVM technique was implemented. The dynamic gestures are defined as a movement of hands in time, where HMM technique was implemented[29].

Static Gesture Recognition – Proposed Method

For the static recognition problem each detected hand’s position and orientation are being treated as a new data point to easily generated multiple samples from the sensors in short time, and treat this as a classification problem. The 3D data yield by the sensor is represented in the sensor’s coordinate system. Therefore, even a small change of the position or orientation of a hand with respect to the location of the sensor can negatively affect the gesture recognition system[29].

SVM Technique - Implementation

The Support Vector Machines (SVMs) were used as a robust classification algorithm. The SVMs were chosen, as there exist a solid mathematical background supporting the simple idea of maximizing the margin between classes. Another advantage is the existence of the open-source library libSVM, which contains an efficient implementation of the SVM in many programming languages. In the original work, SVMS were designed to be able to classify between two classes, but the idea was expanded to utilize the one-vs-all classification to allow differentiating between multiple classes[29].

The problem of classification assumes that each sample consists of set of features that describe a sample that distinguish it from each other. Additionally, each sample has a known or unknown label to define the membership of the sample to the classification class. The samples with known labels can be used to train the classification system. The trained classification system is used to predict the labels for the unlabeled samples. In application of gesture recognition the classification is divided into two modes: the training part, or offline, and the recognition part, or online. The static gesture processing flow is presented in the figure below. In the training part, the library is provided with the recorded and labeled samples of static gestures. From those samples, the sets of features are computed for classification purpose. The recognition part assumes to have trained classifier. The recognition part is provided with samples of static gestures without labels, where sets of features are computed and then given as input to the trained classifier. The classifier provides the information of the predicted gesture’s class membership (label) for each input sample[29].

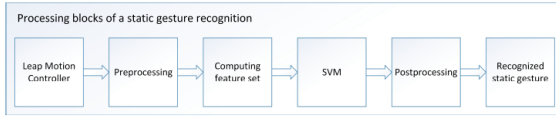
This might be a drawback, as for n classes; it is necessary to train n SVMs, each to classify one class against all samples from the rest of the classes. The efficiency of SVMs depends on correctly choosing the kernel function used to map the separation problem into higher dimension with expectation to achieve a classification problem that can be separated[29]. There exist several kernel functions:

Where, r , and d are kernel parameters. According to the authors of the library, linear kernels perform the best, when used for linearly separable problems, while RBF kernels are the most versatile ones and are recommended for most of the applications[29].

Proposed solution blocks for learning and recognition parts of static gestures recognition problem.

- linear: $K(x_i, x_j) = x_i^T x_j$,
- polynomial: $K(x_i, x_j) = (\gamma x_i^T x_j + r)^d, \gamma > 0$,
- radialbasisfunction(RBF): $K(x_i, x_j) = \exp(-\gamma \|x_i - x_j\|^2), \gamma > 0$,
- sigmoid: $K(x_i, x_j) = \tanh(\gamma x_i^T x_j + r)$,

Figure 19: Proposed Solution



While the presented approach can be treated as state-of-the-art approach it still cannot be used without defining proper feature sets for gesture recognition. The literature suggests computing a set of features invariant to wanted transformations, which can fully distinguish between different classes[29].

Detection of dynamic gestures - Proposed methods

In the dynamic gesture recognition process the input data consist of a list of consecutive positions and orientations of hand and fingers. The important factor for recognition is the time dependencies between sequences of hand and finger positions. Also, gestures performed slower and faster should be recognized as the same dynamic gesture. The proposed solution utilizes parts of the solution used for recognition in the static gestures. The feature set describes each frame of the captured data. The set of features for each frame is then processed by the Hidden Markov Model scheme[29].

Hidden Markov Model

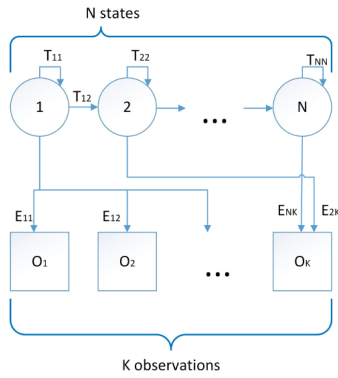
A HMM can be considered as a finite, N -element set of states associated with the probability of distribution. A Hidden Markov Model is a model of a system with the Markov property. The Markov property means that the next future state depends only on the previous state and probability of transition between those states[29]. The transitions between states are represented by the transition probabilities usually stored in $N \times N$ matrix T . In every state, one of the observations from the finite, K -element observation set can be generated with observation probability usually represented by the $N \times K$ emission matrix E . The finite set of all possible observation is called the alphabet. The HMM also consists of a N - element vector of initial state probabilities of π the hidden starting state. Each HMM can be fully defined by the (T, E, π) .

The example HMM can be seen in Fig. 6.1. The HMM in the figure consists of N states $\{1, 2, 3, N\}$ and K possible observations $\{O_1, O_2, O_k\}$. The probabilities T_{ij} define the transition probability from state i - th to state j -th. The probabilities E_{ij} define the observation probability of generating j observation while being in state i [29]. The non-zero transition and emission probabilities are represented in Fig 6.1 by arrows.

The HMM can be understood as a directed graph with two types of weighted vertices. This way, each state is represented by one type of vertices while observations can be shown as second type of vertices. There are no edges between vertices representing observations. The probability values for edges are stored in T and E matrices[29].

There are three main algorithms used with the HMMs:

Figure 20: HMM LM Algorithm



Forward-Backward algorithm
Viterbi algorithm
Baum-Welch algorithm

Forward-Backward Algorithm

The Forward-Backward algorithm finds the posterior probability of given states given the set of observations. Given the set of N hidden state variables $X = \{X_1, X_2, \dots, X_N\}$ and a sequence of t observations $o_1 : t = \{o_1, o_2, \dots, o_t\}$, the algorithm computes probability of state given the observed sequence $P(X_i | o_1 : t)$. The algorithm utilizes performs three steps in a loop:

1. Computing forward probabilities
2. Computing backward probabilities
3. Computing smoothed values

The forward pass phase for all states $i = \{1, \dots, N\}$ computes probability $P(X_i | o_1 : k)$, where k is smaller than t that represents the probability of ending up in state X_i after first k observations. The backward pass computes probability $(P_{o_k+1:t} | X_i)$, which are the probabilities of observing the rest of the observations from the state X_i [29]. The smoothing part uses Bayes rule to compute the probability of state X_i given the whole observation sequence:

$$P(X_k | o_1 : t) = P(X_k | o_1 : k, o_{k+1} : t) \propto P(o_{k+1} : t | X_k) P(X_k | o_1 : k).$$

The time complexity of this algorithm is $O(N^2T)$, where T is the length of observation sequence and N is the number of possible states[29].

Viterbi Algorithm

The Viterbi algorithm is used to find the most likely sequence of hidden states that best explain the set of observations. The set of those states is called the Viterbi path[29].

The solution can be understood as finding a path with maximum probability $\arg \max_x E^X(VT, x)$. The time complexity of the algorithm is $O(N^2T)$, where N is the number of states and T is the observation length and is equal to the complexity of the Forward-Backward algorithm[29].

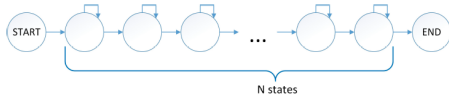
Baum-Welch Algorithm

The Baum-Welch algorithm is the algorithm used to find the unknown parameters of the HMM. For each training example, the algorithm computes the state transition probabilities, observation probabilities in each state and initial probabilities, which maximize the likelihood of observed sequence. Having the set of sequences of observations, the algorithm can be used to train HMM to detect the sequences similar to the ones used in learning process. The algorithm uses the results obtained by the Forward-Backward algorithm to iteratively update the probabilities in the emission and the transition matrices. The training process is stopped after the desired level of a convergence is reached[29].

Structure of the HMM

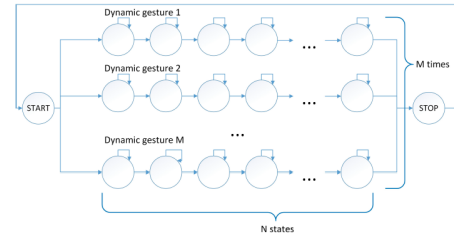
In the dynamic gesture recognition task, we adopted a structure of HMM where each state has nonzero transition probabilities to itself and to the next state in the sequence. The proposed structure is presented in Fig. 6.2 and was proposed by Yang et al. The states can be understood as the phases of hand movement that happen when a wanted gesture is performed. The self-transitions are used to model the different speeds of the gestures and thus allow achieving a more robust system. This structure after training process can be used to measure the probability that the dynamic gesture occurred given the set of observations[29].

Figure 21: HMM Structure



Having the single dynamic gesture recognition problem modeled as the sequence of N states in which k -th state is connected by the edges to the k and $(k+1)$ state and to all observations.

Figure 22: HMM Structure 2



The problem of distinguishing M gestures translates to the problem of computing probabilities for M sequential graphs. The problem of finding whether and what dynamic gesture occurred is the problem of finding the probabilities from each of the M HMMs. Alternatively, the M HMMs can be combined into one HMM where those single HMMs are treated as parallel paths. The structure of proposed single HMM is presented in Fig. 6.3. In this structure, the recognition process is a process of finding, which of the parallel paths is the most probable[29].

Structure of the HMM's states and non-zero transitions used to simultaneously detect one of M dynamic gestures. HMM Observation from Leap Motion Data One of the problems when designing the system is a problem of correctly preparing the set of observations from the Leap Motion data. This poses a challenge as the raw data from Leap Motion is high dimensional[29]. This is the reason, why it is needed to reduce the dimensionality of the observation of the single hand in the predefined moment in time. To do this, unsupervised clustering algorithms were introduced. For this task we examined 3 methods:

Chinese whispers

Girvan-Newman clustering

K-means with additional methods to determine the number of clusters

Chinese Whispers

Chinese whispers algorithm is an efficient randomized graph-clustering algorithm that is time-linear in the number of edges. The algorithm was proposed in work by Biemann and thoroughly examined in Natural Language Processing problems. The main advantage of the algorithm is the ability to independently determine the number of classes. In tests, we used the implementation available in dLib-ml library, which provides multiple implementations of machine learning algorithms[29].

Girvan-Newman Clustering

Girvan-Newman clustering is a hierarchical algorithm used to detect clusters in a data represented as a graph. The algorithm progressively removes edges from the original network until it reaches the maximal number of iterations or an error condition is met. An adjacency matrix is used to represent the edges between data, where edge value is the similarity between two data points. The algorithm iteratively calculates the eigenvalues of a created matrix using the power iteration method. In case of a complete graph, this algorithm is relatively slow[29].

K-means Clustering

Another proposed approach is k-means clustering algorithm. The idea for the algorithm comes from polish mathematician Hugo Steinhaus, and MacQueen presented the first practical application[29]. The algorithm initially randomly selects k starting points and iteratively performs two steps utilizing the idea of expected-maximization:

1. For each data sample the algorithm calculates the distances to k centroids and labels the samples with the label of the centroid with the smallest distance.
2. For each class, the algorithm recalculates centroids position to the averaged position of all samples belonging to class.

The algorithm stops after the maximum number of iterations or when the change between two consecutive iterations is smaller then defined value. Unfortunately, the algorithm requires prior knowledge of the number of expected classes. Nevertheless, there exist heuristics methods aiding in correctly choosing the number of classes. One of those methods is plotting the sum of squares error (SSE) between elements within each class against the number of chosen classes. The plot is then visually inspected and the fall in error is sought. The argument for which the drastic error drop is observed is believed to be the correct number of classes in the provided data[29].

Another, more formal approach introduces the measure of dissimilarity and is called Silhouette width. Declaring $a(i)$ as the average dissimilarity measure to other samples in the same cluster and $b(i)$ as the lowest average dissimilarity measure to samples in the other clusters, which i is not a part of[29]. The measure of each cluster corrected is computed by:

Figure 23: Eq.LM

The big value of $s(i)$ implies good clustering, while small value of $s(i)$ suggests that the number of classes was not properly chosen. The $s(i)$ values are used to find the proper number of clusters[29].

$$s(i) = \frac{b(i) - a(i)}{\max\{a(i), b(i)\}}$$

HMM Implementation

The solution consists of two parts: offline learning and an online recognition. In learning part, the raw data is preprocessed and then the feature set is extracted. Those features are extracted for all recorded positions in training data. Then, one of the unsupervised clustering algorithms is used to find the clusters in the data and then classify those feature sets into observation clusters. Computed sequences of observation classes represent each dynamic gesture as a series of one-dimensional observations. For each dynamic gesture, the corresponding HMM is learnt using the Baum-Welch algorithm on the sequence of observations. The Baum-Welch algorithm computes the matrices that maximize the probability for one sequence of observations. The problem of training on multiple training data was encountered. Using the idea presented by Petrushin, where each matrix trained by one training example is incorporated into HMM's transition matrix using addition with a learning rate μ [29]. Denoting the transition matrix of HMM by T and learnt transition matrix T_{new} , it can be written:

$$T = (1 - \mu)xT + \mu xT_{new} :$$

Same idea is applied to the emission matrix E and learnt emission matrix E_{new} :

$$E = (1 - \mu)xE + \mu xE_{new} :$$

The learning rate μ is a value between 0 and 1. Smaller learning rates results in slower learning, but bigger values may result in no convergence of learning. Finding the correct learning rate is a task of one of the experiments. Another problem encountered using this approach is the fact, that matrices no longer represent the probability, as the total values in rows do not add to 1:0. Therefore, each row of new matrices is normalized to 1:0. When dealing with multiple training examples, the learnt solution may adapt to the training samples, but perform poorly on the testing set. The problem is known as over-fitting to the training samples. To deal with potential problem, the k-fold cross validation was utilized in a similar approach as in the static gestures. To complete the training process, each trained HMM is combined into a single recognition structure[29].

In the recognition, the raw data from Leap Motion is preprocessed, and then each frame is labeled accordingly to the classes learnt by the unsupervised clustering algorithm. The next step includes providing the set of observations to HMMs and running the Viterbi algorithm. The Viterbi algorithm computes a probability of the most likely sequence. Then, the maximum probability for each HMM is calculated. If the maximum probability is above threshold, the gesture is assumed to be correctly recognized with the label of the HMM with the maximum probability sequence[29].

4 How Can Automatic Sign Language be Improved

Technology such as text-to-speech and speech-to-text has advanced dramatically in the past few years. Therefore, we must focus on the technology that is currently lacking, sign-language-to-text. Because of this reason, the goal should be to use the most advanced body and hands recognition technology, such as Kinect and Leap Motion, to build an automatic sign language translator to translate sign language to text. The idea is to build a fully functioning sign language automatic translator to resolve the interpersonal communication problem, between the not-speaking signers and the speakers communities, by building up a new communication bridge and integrate components such as sign-to-text, text-to-speech, and speech-to-text. In order to achieve this goal, the development of such project would have to go through three important stages. First stage, the Kinect and Leap Motion APIs integration. Second stage, features extraction, features-to-matrices conversion, data loading and training. Last stage, a SVM and two HMMs algorithms implementation to classify the data, according to the signs and its features, and to translate the signs based on two models. The first model would translate ASL to text through segmented signs built within the system, and the second model would translate signs based on a matrix similarity algorithm used to compare the input sign with those in the trained database.

As a result, the Kinect model and the Leap Motion model would be tested based on their capabilities. The Kinect model would be tested with a dataset containing body gestures such as arms, hands, head, and few fingers movements, whereas the Leap Motion model would be tested with a dataset containing hands and fingers movements. It is expected that, when both models get tested in conjunction, the results of the Kinect model will improve dramatically. In short, the Kinect limitations are apparent when tracking the fingers since it can only track the thumb and tip of the hands. Similarly, Leap Motion can only track the hand and fingers of a person. Nevertheless, the fusion of both devices makes a perfect software recognition by being able to track body and face features coordinates with accurate hands and fingers features coordinates.

5 Conclusion

Deaf and hard of hearing communities revolve around sign languages as their natural means of communication. As a minority community, with limited access to heard speech, deaf and hard of hearing people rely on the relatively small numbers of hearing people who learn sign language to be their interpreters. Sign languages have different grammatical structures to spoken languages, which makes translation a problem in most contexts without provision of highly trained interpreters. An estimated 360 million people worldwide suffer from hearing loss, including approximately one-third of all people over the age of 65. Although, other communication methods exist and the number of people who use sign language continues to grow, communication between people who can and cannot hear, or speak, can be difficult. This is a serious challenge for the deaf and hard of hearing community when integrating into educational, social and work activities. Deaf students in U.S. and Canada are taught written English, but their inability to hear spoken English results, in most deaf U.S. high school graduates, reading at a fourth grade level. The goal of this paper was to collect all the most influential events that has shaped the development of automatic sign language translators, whose goals are the recognition of the right to use an indigenous sign language and to empower this community, as it strives for equality in education and employment. As a result, the analysis of multiple built machine translators' software, with the latest technology, could be used to build more effective educational software for deaf and hard of hearing people to improve their language literacy skills. Furthermore, the speed in which the technology (explained in this paper) is growing, the interpersonal communication problem between signers and the hearing community will be resolved soon. In fact, the integration of advanced communication devices, such as Microsoft Kinect and Leap Motion Controllers, are becoming the new communication bridge integrating components for sign, speech, and text processing in their automatic translation systems.

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